

Classical Control of a DIY Reflow Oven: Design, Simulation, and Hardware Validation of a PI + Feedforward Thermal Tracking Controller

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Abstract—Commercial reflow ovens deliver the precise thermal regulation required for surface-mount soldering, but their cost places them out of reach for hobbyists. This project converts a \$50 toaster oven into a working reflow oven by designing, simulating, and deploying a closed-loop temperature controller. The oven is identified as a first-order plus dead-time (FOPDT) plant from four open-loop bump tests, yielding a nominal model $G(s) = 666.5/(395.3s + 1)e^{-29.6s}$. A proportional–integral (PI) controller is then designed using SIMC direct synthesis. A closed-form analysis shows that PI alone cannot meet the 5 °C steady-state tracking requirement on the ramp portions of the reflow profile because the dead time forces the SIMC closed-loop time constant negative. Adding a model-inversion feedforward term derived from the same plant model and advancing the reference by the dead time θ recovers tracking. Simulink simulation of the PI + feedforward architecture produces near-zero tracking error. Deploying the controller on the physical oven exposed a non-stationary dead time (collapsing from 29.6s to roughly 5s once the cavity warms up) and motivated a re-tuning step in which the proportional gain was scaled 5× and the design dead time reduced. The final hardware controller holds tracking error within $\pm 5^\circ\text{C}$ and remains stable under a 14 % plant-gain shift.

I. INTRODUCTION

Surface-mount technology (SMT) is the dominant assembly method for modern printed-circuit boards. Reliable SMT soldering depends on driving the assembly through a tightly specified reflow temperature profile, with controlled ramp rates, a soak plateau for flux activation, and a reflow ramp where the solder melts. Commercial reflow ovens deliver this regulation but typically cost upwards of \$500, which is out of budget for most hobbyists and students.

A toaster oven, in contrast, can be acquired for under \$50 but provides only crude on/off thermostatic regu-

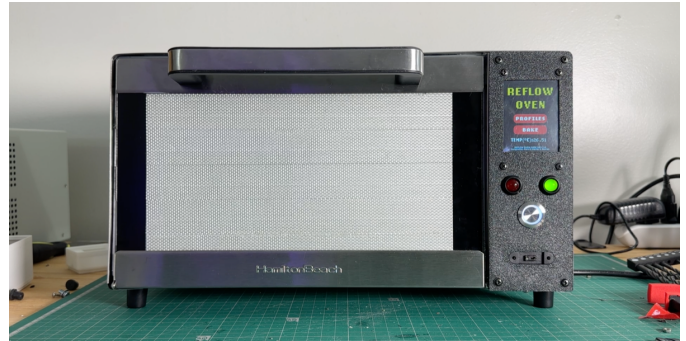


Fig. 1. The DIY reflow oven hardware. A modified countertop toaster oven with an RTD probe inside the cavity, a microcontroller-driven SSR for the heater, and a control panel exposing profile selection, mode, and a temperature readout.

lation – nowhere near the accuracy required for solder paste reflow. The classical control techniques applied here – system identification, direct-synthesis controller tuning, and feedforward augmentation – follow the methods taught in the EECE 5580 course material [1]. The gap between the two can be closed with control. This project takes a previously hardware-modified toaster oven (contains a solid-state relay, an RTD temperature probe, and a microcontroller running the heater PWM loop) and replaces its manually tuned controller firmware with a designed PI + feedforward controller capable of tracking the lead-free reflow profile.

The control problem is non-trivial because the oven exhibits significant transport delay between heater energization and sensor response, due to the thermal mass of the oven. Such plants are well-modeled as first-order plus dead-time (FOPDT) systems, and the dead time imposes a fundamental limit on achievable closed-loop bandwidth that ordinary PI tuning cannot circumvent. A second motivation is personal: the oven is part of the

author's home electronics workbench, so a successful controller upgrade has direct, measurable benefit beyond the academic exercise.

This report differs from the original project proposal in its controller-design path. The proposal anticipated Cohen–Coon PI tuning and possibly a Smith Predictor; in practice, SIMC direct synthesis combined with model-inversion feedforward proved more tractable and is the approach reported here.

II. OBJECTIVES AND SCOPE

The project deliverables are organized as three goal levels of increasing ambition.

Level 1 – System Identification: Capture the open-loop step response of the oven and derive an FOPDT transfer function from the data. Success is defined as a fitted model whose simulated step response visually matches the measured data.

Level 2 – Controller Design and Simulation: Design a closed-loop controller that tracks the preheat ramp, soak, and reflow segments of the lead-free reflow profile in simulation. Success is defined by a steady-state tracking error no larger than $\pm 5^\circ\text{C}$ during the soak segment.

Level 3 (Extra Credit) – Hardware Deployment: Implement the controller onto the oven's microcontroller, run a full reflow cycle, and verify that the simulation prediction translates to physical hardware.

Assumptions and Out-of-Scope Items

The work assumes (i) the heating element is powerful enough to sustain a positive ramp of 1°C/s to 3°C/s up to a peak of 250°C , and (ii) the RTD provides accurate temperature readings with negligible delay relative to the cavity dead time. Active cooling control, leaded-solder paste profiles, and adaptive identification across additional duty cycles are out of scope.

III. METHODS

A. Plant Modeling

The oven is modeled as a first-order plus dead-time process,

$$P(s) = \frac{K}{\tau s + 1} e^{-\theta s}, \quad (1)$$

where K is the steady-state gain in $^\circ\text{C}$ per normalized PWM input, τ is the time constant, and θ is the transport (dead) time between PWM duty cycle change and sensor response [2].

B. System Identification (Level 1)

Four open-loop bump tests were performed at PWM duty-cycle counts of 26, 64, 102, and 255 (out of 255), corresponding to normalized inputs $u_{\text{step}} \in \{0.10, 0.25, 0.40, 1.00\}$. For each test the oven was driven from ambient at the fixed duty cycle and the RTD temperature was logged until the response had clearly reached steady state. The FOPDT parameters were extracted using a Python script generated with the assistance of Claude [3], implementing the methods in [2], [4], [5]. The results are summarized in Table I.

TABLE I
FOPDT PARAMETERS EXTRACTED FROM OPEN-LOOP BUMP TESTS AT FOUR DUTY CYCLES. THE DUTY 0.40 ROW WAS SELECTED AS THE NOMINAL DESIGN MODEL.

Duty	u_{step}	K ($^\circ\text{C}$)	τ (s)	θ (s)
26	0.10	913.0	585.7	52.7
64	0.25	760.0	482.9	40.0
102	0.40	666.5	395.3	29.6
255	1.00	404.8	193.7	19.5

The duty-0.40 operating point was chosen as the nominal model because it sits near the middle of the reflow profile's average PWM demand, balancing the more delay-dominated low-duty regime against the more aggressive full-power regime. The selected nominal plant is therefore

$$G(s) = \frac{666.5}{395.3s + 1} e^{-29.6s}. \quad (2)$$

The wide variation of K and θ across operating points (both fall by roughly a factor of two from low to high duty) means that the controller must be robust to substantial parameter drift.

C. PI Controller Design via SIMC Direct Synthesis

The reflow profile demands tracking of ramp inputs. With a Type-0 plant such as (1), a proportional-only controller yields infinite steady-state error to a ramp; adding integral action raises the system to Type 1 and gives a finite ramp error

$$e_{\text{ss}} = \frac{R}{K K_i}, \quad (3)$$

where R is the ramp rate. A PI controller $C(s) = K_p + K_i/s$ is therefore the minimal structure that can in principle hit the steady-state specification.

The controller gains are derived using Skogestad's SIMC direct-synthesis rules [6]. Choosing the desired closed-loop response $T_{cl}(s) = \frac{1}{\tau_c s + 1} e^{-\theta s}$ and using the

first-order Taylor expansion $e^{-\theta s} \approx 1 - \theta s$, solving $C(s) = T_{cl}/[P(s)(1 - T_{cl})]$ yields

$$K_p = \frac{\tau}{K(\tau_c + \theta)}, \quad K_i = \frac{1}{K(\tau_c + \theta)}. \quad (4)$$

The maximum ramp rate of the lead-free profile is $R = (150 - 25)/(90 - 0) \approx 25/18$ °C/s. Substituting (4) into (3) and demanding $e_{ss} < 5$ °C gives a constraint on the closed-loop time constant,

$$\begin{aligned} R(\tau_c + \theta) &< 5^\circ\text{C}, \\ \tau_c + \theta &< \frac{5}{R} = \frac{5}{25/18} = \frac{18}{5}, \\ \tau_c &< \frac{18}{5} - \theta. \end{aligned} \quad (5)$$

For the nominal model in (2), $\theta = 29.6$ s already exceeds $18/5 = 3.6$ s, so (5) forces $\tau_c < -26$ s, which is physically impossible. *The PI controller alone cannot satisfy the steady-state tracking specification for this plant.*

D. Feedforward Augmentation

The contradiction is resolved by adding a feedforward term that takes the bulk of the control effort away from the integrator. For a ramp reference $r(t) = Rt$ ascending from the current temperature, the steady-state input required to follow the ramp through the FOPDT plant is found by inverting the plant in the low-frequency limit, giving

$$u_{ff}(t) = \frac{\tau R + r(t)}{K}, \quad (6)$$

where $r(t)$ is the current commanded reference temperature and R the local ramp rate. The reference fed to the feedback path is additionally advanced by θ seconds so that, by the time the resulting heat reaches the sensor, the response is aligned with the desired profile.

The PI loop is left in place to correct residual model error and small disturbances; the SIMC gains in (4) are kept as the starting design point. The resulting Simulink architecture is shown in Fig. 2; the closed-loop simulations and the discrete controller used on hardware were both built and exercised in MATLAB/Simulink [7].

IV. RESULTS AND DISCUSSION

A. Level 1 – System Identification

The four-point bump-test campaign of Table I satisfies the Level 1 deliverable: the duty-0.40 FOPDT model in (2) reproduces the corresponding measured step response visually, and the parameter spread across operating points provides a quantitative estimate of the model uncertainty the controller must absorb.

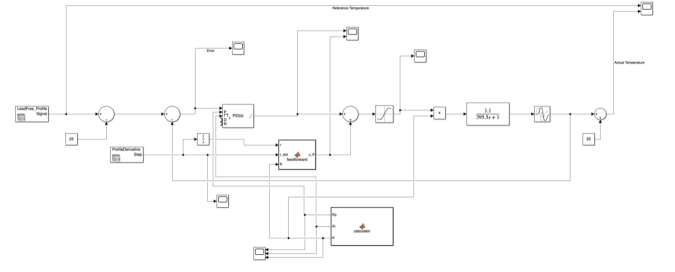


Fig. 2. Simulink implementation of the PI + feedforward controller. The feedforward block computes u_{ff} from (6); the PI block trims residual error. A scalable plant-gain block on the feedback path is used to stress-test robustness to parameter drift.

B. Level 2 – Closed-Loop Simulation

With the SIMC-tuned PI gains in (4) and the feedforward law (6) in place, the simulated response is essentially indistinguishable from the reference profile. Fig. 3 overlays the reference and actual cavity temperature; the two traces are visually coincident through the entire preheat ramp, soak, and reflow segments.

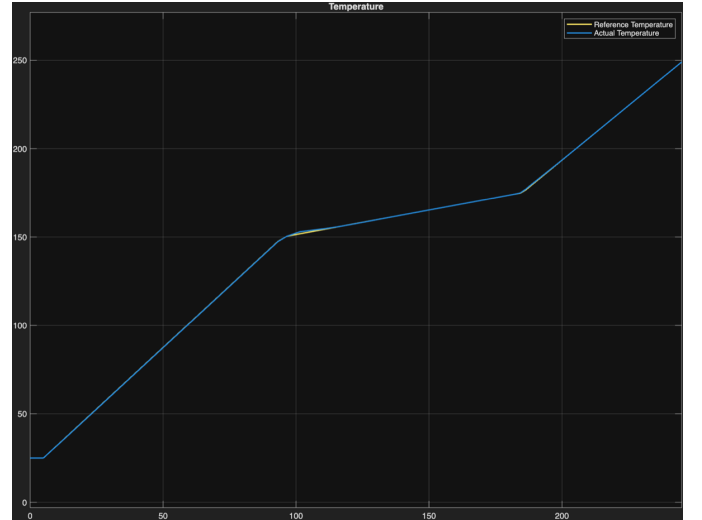


Fig. 3. Simulated closed-loop tracking of the lead-free reflow profile. Reference (yellow) and actual cavity temperature (blue) overlap to within line width across the full profile.

Summary of Levels 1–2. The bump-test delivered a useable FOPDT model, and the PI + feedforward architecture meets the 5 °C steady-state tracking specification with significant margin in simulation. With nominal parameters the worst-case observed tracking error is sub-degree.

V. EXTRA CREDIT – LEVEL 3 HARDWARE DEPLOYMENT

The simulated controller was implemented oven’s microcontroller. This section documents what happened on hardware, why the simulation prediction did not transfer cleanly on the first attempt, and how the controller was re-designed to meet the specification on the physical system.

A. First Hardware Run

Fig. 4 shows the temperature trace and tracking error from the first end-to-end hardware run with the nominal SIMC gains. The error exceeded $\pm 15^\circ\text{C}$ – roughly three times the spec – particularly during the preheat ramp and again at the entry to the reflow stage.

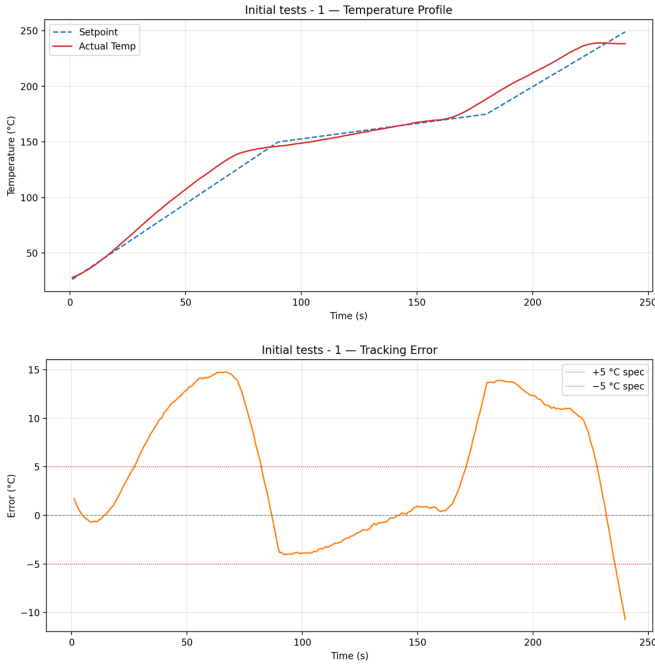


Fig. 4. First hardware run with the simulation-tuned controller. Top: setpoint vs. measured oven temperature. Bottom: tracking error. Peaks exceed $\pm 15^\circ\text{C}$, well outside the 5°C specification.

B. Root-Cause Analysis

Three issues were identified, in decreasing order of impact:

a) *Non-stationary dead time:* The dead time θ is not constant across the reflow cycle. The value $\theta \approx 29.6\text{s}$ extracted from the ambient-start bump test reflects the cold-start transport delay through a cold oven. Once the oven is preheated and the internal air mass is already hot, the effective θ falls to roughly 5 s. The fixed 29.6 s reference advance therefore over-corrects for most of

the reflow profile, pushing the actual oven temperature ahead of the setpoint. Auxiliary tests with the preheat-stage dead time tweaked separately (Fig. 5) confirmed the same trend.

b) *Feedforward sensitivity to plant gain:* Because u_{ff} is doing the bulk of the control work, *any* mismatch between the design K and the actual operating-point gain feeds directly into a tracking error that the PI loop must clean up. With a proportional gain set by the SIMC formula at small magnitude, the PI loop simply does not have enough authority to correct this error within the duration of a ramp.

c) *PI proportional gain too low:* The SIMC choice of K_p in (4) is conservative when the plant model is trustworthy; here, with both the dead-time mismatch and the gain mismatch loading the PI loop with non-trivial residual error, K_p needs to be considerably more aggressive than the nominal SIMC value.

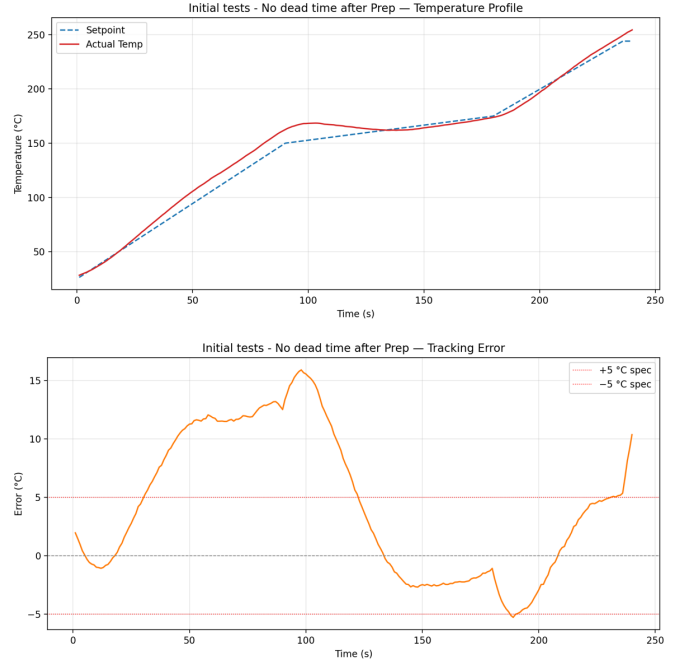


Fig. 5. Hardware run after removing the dead-time advance past the preheat stage, used to confirm that θ is not a single fixed number across the cycle. Mid-cycle behavior improves but cold-start tracking degrades.

C. Controller Re-design

Rather than build a gain-scheduled or adaptive controller, the solution was to re-design once more in simulation against a plant model that better represents the warm-cycle behavior. The design dead time was reduced from 29.6 s to 5 s, the SIMC structure was preserved, and

the proportional gain was treated as a free parameter to be swept. Before re-flashing the firmware, the new controller was stress-tested in simulation against a $1.1\times$ plant-gain perturbation (Fig. 6); even with this 10% gain mismatch the simulated tracking error stays comfortably within $\pm 5^\circ\text{C}$, validating the re-tuned design as worth deploying.

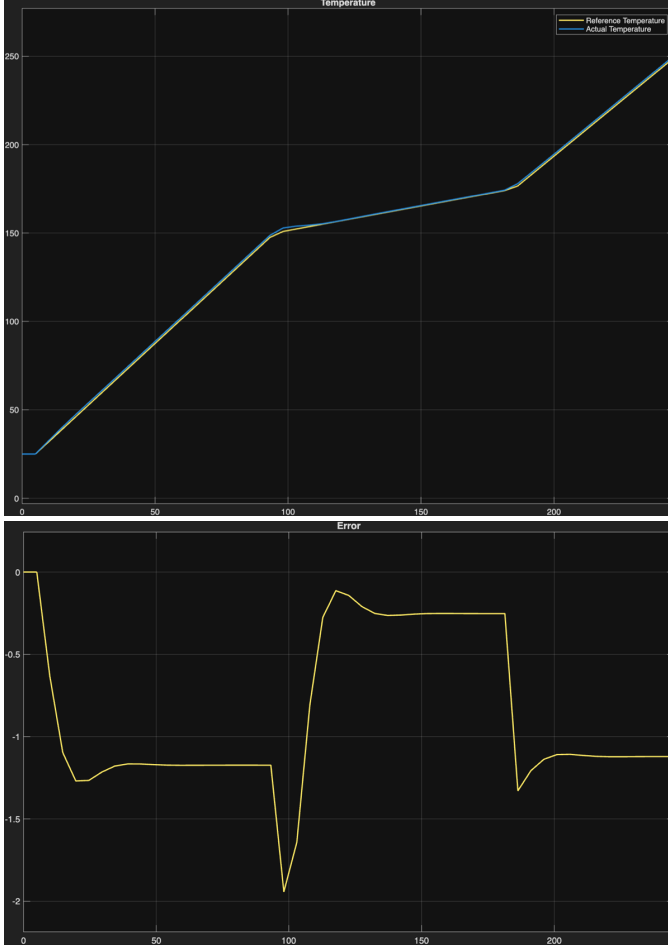


Fig. 6. Simulated robustness check on the re-designed controller (design $\theta = 5$ s): plant gain scaled to $1.1K$. Tracking error stays inside $\pm 5^\circ\text{C}$.

D. Hardware Re-tuning Sweep

Three values of K_p were trialed on the physical oven: $1\times$, $3\times$, and $5\times$ the nominal SIMC value, all at design gain $K = 666.4$. Figs. 7–9 show the resulting temperature traces and error signals.

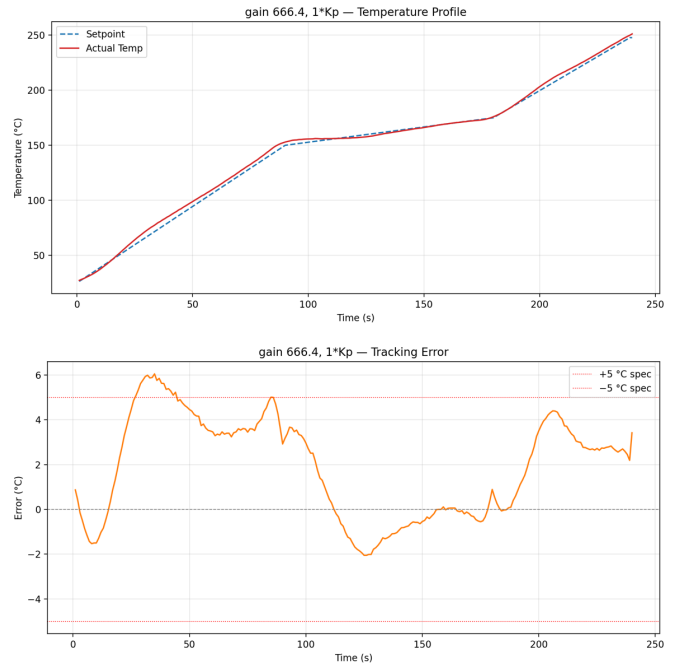


Fig. 7. Hardware result with $1\times K_p$ (re-tuned design, $\theta_{\text{des}} = 5$ s). Better than Fig. 4 but error still drifts outside $\pm 5^\circ\text{C}$ on the steeper ramps.

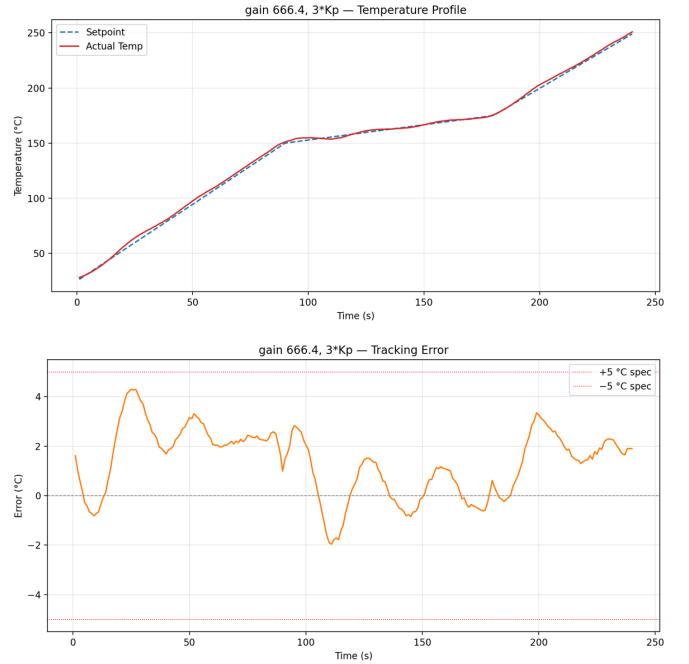


Fig. 8. Hardware result with $3\times K_p$. Error band narrows further; small oscillations begin to appear in the soak plateau.

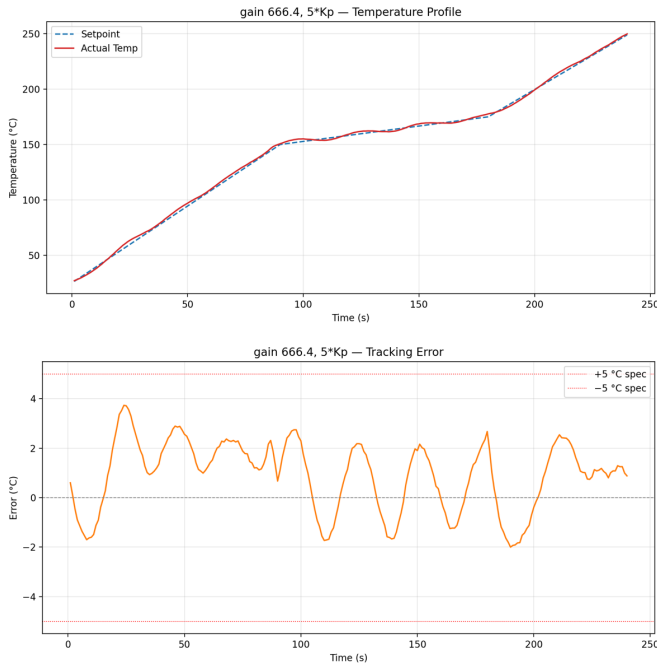


Fig. 9. Hardware result with $5 \times K_p$. Tracking error remains within $\pm 5^\circ\text{C}$ for the entire profile; modest oscillations in the soak are an acceptable trade for meeting the steady-state spec.

The progression $1 \times \rightarrow 3 \times \rightarrow 5 \times$ illustrates the classic trade-off between PI aggressiveness and oscillatory behavior. At $5 \times K_p$ the tracking error envelope is within the 5°C specification across the entire profile, while the small soak-stage oscillations stay well under any thermal-damage threshold.

E. Robustness Cross-Check

To confirm that the $5 \times K_p$ tune is not a brittle local optimum, a final hardware run was performed at the *higher-gain* operating point $K = 759.9$ – a 14 % departure from the design $K = 666.4$. The result is shown in Fig. 10: the controller remains stable and tracking error stays within the same $\pm 5^\circ\text{C}$ band. This validates the controller’s robustness to the kind of operating-point drift visible in Table I.

F. Limitations

A few limitations should be noted. First, the dead time is treated as piecewise-constant by the controller (one value at design time) when in reality it varies continuously with cavity temperature; a gain-scheduled or estimated- θ scheme would likely produce tighter tracking. Second, the current implementation lacks active cooling, so the cooldown segment of the reflow profile is open-loop. Third, the re-tuning was performed against a single re-design point ($\theta = 5\text{ s}$) without a formal

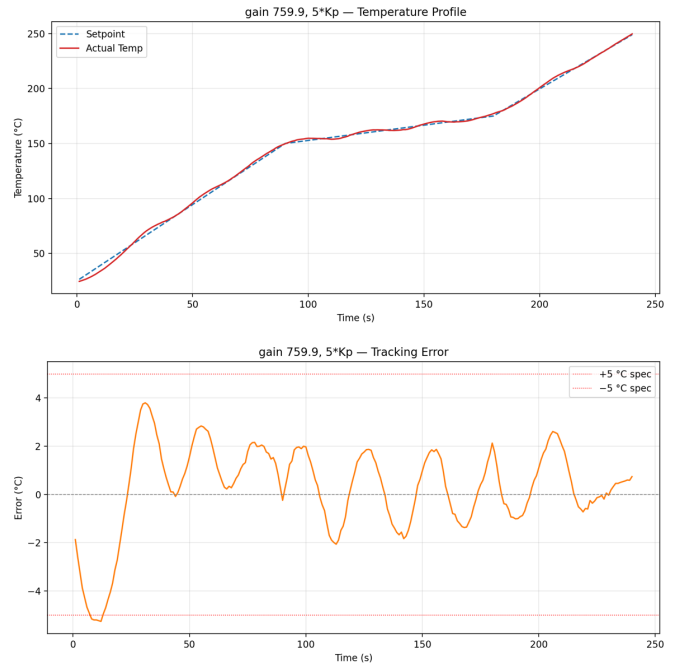


Fig. 10. Robustness cross-check at $K = 759.9$ (14 % above design $K = 666.4$), with $5 \times K_p$. Tracking error remains within the 5°C specification.

sensitivity sweep on θ ; this was acceptable given the goal of meeting the 5°C spec but would benefit from more systematic study.

VI. CONCLUSION AND FUTURE WORK

The project demonstrates an end-to-end classical-control workflow on a real, low-cost piece of hardware: identify the plant from open-loop bump data, design a PI controller via SIMC synthesis, recognize the analytic infeasibility of meeting the ramp-tracking specification with PI alone, augment with a model-inversion feedforward, validate the design in Simulink, and then iterate on the design once the first hardware deployment exposed model assumptions that did not hold across the full reflow cycle. After re-designing against an effective warm-cycle dead time and scaling the proportional gain $5 \times$, the physical oven tracks the lead-free reflow profile with error inside $\pm 5^\circ\text{C}$ and remains stable under a 14 % plant-gain shift.

The natural next steps focus on extending the system, not replacing it. First, closing the loop on the cooldown segment will require adding active cooling (a small fan or vent actuator) so the controller can track the descending portion of the reflow profile rather than letting the oven coast. Second, the system-identification step is currently a manual procedure; automating the bump-test routine

on the microcontroller – so the oven can re-extract its own K , τ , θ on demand – would let the controller adapt to changes in heater age, ambient temperature, or board loading without operator intervention. Third, the firmware will be extended to support a leaded-solder paste profile in addition to the lead-free profile demonstrated here, so the same hardware can serve both contemporary and legacy assembly work.

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